

PFC Boost Inductor Engine — Equation Reference

Step-by-step equations for every quantity computed by **pfc_inductor_engine.py** (v1.0.0, schema 1.0) and the **SimAgentField** browser viewer (v14). Transcribed from the code; companion to DATA_DICTIONARY.md and PowderCore_Magnetic_Calc_Standard_v1.

Conventions. Distributed-gap powder toroid, CCM boost PFC, n_{ph} -phase interleaved. Geometry in mm; flux density in T; bias field H in oersted (Oe); Steinmetz frequency in kHz ; resistance in Ω ; power in W; temperature in $^{\circ}\text{C}$. $\theta \in (0, \pi)$ is the line half-cycle angle; the engine evaluates all θ -dependent quantities on a 40 001-point grid and integrates by the trapezoidal rule. Subscript “single” = one core of the stack. Each quantity carries a provenance/tier: analytic-T1, FEA-T2, measured-T3; solved or measured data, when present in the package, replaces the analytic expression for that quantity only (precedence: measured > FEA > analytic).

S0 — Physical constants (the only numbers owned by the engine)

(E1) $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

- μ_0 — permeability of free space

(E2) $1 \text{ Oe} = 79.577 \text{ A/m}$

- oersted-to-SI conversion for the bias field H

(E3) $\rho_{20} = 1.724 \times 10^{-8} \Omega \cdot \text{m}$, $\alpha_{Cu} = 0.00393 \text{ K}^{-1}$

- ρ_{20} — copper resistivity at 20 $^{\circ}\text{C}$ (package-overridable)
- α_{Cu} — copper temperature coefficient (package-overridable)

Note: Every material, geometry, wire and operating value is injected by the package; none is built in.

S1 — Unbiased inductance and tolerance band

(E4) $L_0 = A_{L,\text{single}} \cdot n_{\text{stk}} \cdot N^2 \times 10^{-9} [\text{H}]$

- $A_{L,\text{single}}$ — inductance factor of ONE core at zero bias [nH/turn^2]
- n_{stk} — number of stacked cores; N — turns

(E5) $L_{0,\text{min}} = L_0(1 - \text{tol}_{AL})$, $L_{0,\text{max}} = L_0(1 + \text{tol}_{AL})$

- tol_{AL} — catalog A_L tolerance fraction (e.g. 0.08 for $\pm 8\%$)

Note: The tolerance derate applies ONLY when L(H) is analytic. A solved FEA table or measured L(I) is the actual inductance and is used un-derated (parity rule, both engine and viewer).

S2 — DC-bias field and permeability retention

(E6) $H = 0.4\pi \cdot N \cdot I / l_{e,\text{cm}} [\text{Oe}]$

- I — instantaneous inductor current [A]; $l_{e,\text{cm}} = L_{e,\text{mm}}/10$ — magnetic path length [cm]

(E7) $k(H) = 1 / (k_0 + k_1 \cdot H^p) / 100 [-]$

- $k(H)$ — per-unit permeability retention (catalog DC-bias fit; $k_0 \approx 0.01$ gives 100 % at $H=0$)
- k_0, k_1, p — material retention coefficients from the package

(E8) $L(H) = k(H) \cdot L_0$ (analytic T1) | $L(H) = \text{interp}(H; \text{table})$ (FEA T2 / measured T3)

- table — fields.inductance{H[Oe], L[μH]} or material.measured.inductance

Note: Measured L given versus bias current is converted point-by-point with E6: $H_i = 0.4\pi \cdot N \cdot I_i / l_{e,\text{cm}}$.

S3 — Operating-point currents (per spec corner { V_{in} , P_{out} , η , PF})

(E9) $P_{in} = P_{out} / \eta$

- η — converter efficiency at this corner (per unit)

(E10) $V_{pk} = \sqrt{2} \cdot V_{in}$

- V_{in} — line RMS voltage [V]

$$(E11) \quad I_{in,rms} = P_{in} / (V_{in} \cdot PF), \quad I_{in,pk} = \sqrt{2} \cdot I_{in,rms}$$

- PF — power factor at this corner (per unit)

$$(E12) \quad I_c = I_{in,pk} / n_{ph}$$

- I_c — per-phase crest (low-frequency peak) current; n_{ph} — interleaved phases

Note: Browser map form: when only $maps.etaByVin$ is given, $\eta \cdot PF$ is taken as the map value (the PRODUCT) with $P_{out} = load \cdot P_{rated}$; `validate()` cross-checks the two forms.

S4 — Line-cycle waveforms over $\theta \in (0, \pi)$

$$(E13) \quad v_{in}(\theta) = V_{pk} \sin\theta, \quad D(\theta) = \text{clip}(1 - v_{in}(\theta)/V_{out}, 0, 1)$$

- $D(\theta)$ — boost duty ratio (CCM); V_{out} — DC output voltage
- $D_{pk} = 1 - V_{pk}/V_{out}$ — duty at the line crest

$$(E14) \quad i_{avg}(\theta) = I_c \sin\theta$$

- i_{avg} — switching-period-average inductor current (sinusoidal PFC reference)

$$(E15) \quad \Delta I_{pp}(\theta) = v_{in}(\theta) \cdot D(\theta) / (L(H(\theta)) \cdot f_{sw})$$

- ΔI_{pp} — peak-to-peak switching ripple; L evaluated at the local bias $H(\theta) = 0.4\pi \cdot N \cdot i_{avg}(\theta) / I_{e,cm}$ (E6)
- f_{sw} — switching frequency [Hz]

$$(E16) \quad i_{pk}(\theta) = i_{avg}(\theta) + \Delta I_{pp}(\theta)/2$$

- i_{pk} — instantaneous switching-peak current

$$(E17) \quad B_{ac,pk}(\theta) = v_{in}(\theta) \cdot D(\theta) / (2 \cdot N \cdot A_e \cdot f_{sw})$$

- $B_{ac,pk}$ — half of the peak-to-peak AC flux swing [T]
- $A_e = A_{e,single} \cdot n_{stk} \times 10^{-6}$ — total effective area [m²]

$$(E18) \quad B_{dc}(\theta) = L \cdot i_{avg}(\theta) / (N \cdot A_e), \quad B_{mean}(\theta) = B_{dc}(\theta) + B_{ac,pk}(\theta)$$

- B_{dc} — bias flux density at the mean magnetic path; B_{mean} — instantaneous peak at the mean path

S5 — Peak flux density and saturation margin

$$(E19) \quad B_{max} = \max_{\theta} [L(H(\theta)) \cdot i_{pk}(\theta) / (N \cdot A_e)] \quad [T, \text{mean path}]$$

- the worst instantaneous mean-path peak over the half line cycle

Note: Acceptance basis (parity rule): B_{max} is checked against the upstream limit $B_{max,T}$ or $(1 - \text{sat_margin_min}) \cdot B_{sat}$; the verification golden numbers use this mean-path basis.

$$(E20) \quad \text{crowd} = r_{mean} / r_{in} \quad (\text{analytic } 1/r) \quad | \quad \text{crowd} = \text{interp}(r_{in}; \text{fields.flux.radial})$$

- $r_{in} = ID/2$, $r_{mean} = (ID/2 + OD/2)/2$ — inner and mean radii [mm]

$$(E21) \quad B_{inner} = B_{max} \cdot \text{crowd}$$

- B_{inner} — crowded inner-radius peak flux (toroid $1/r$ concentration)

Note: Checked only if upstream supplies `acceptance.Binner_max_T`.

$$(E22) \quad \text{margin}_{sat} = (B_{sat} - B_{max}) / B_{max} \times 100 \%$$

- B_{sat} — material saturation flux density [T]

S6 — Copper geometry and DC resistance

$$(E23) \quad A_{str} = (\pi/4) \cdot d_{str}^2, \quad A_{Cu} = n_{par} \cdot n_{str} \cdot A_{str} \quad [\text{mm}^2]$$

- d_{str} — BARE strand diameter [mm]; n_{str} — strands per bundle; n_{par} — parallel bundles

$$(E24) \quad w_{core} = (OD - ID)/2, \quad h_{stk} = HT_{single} \cdot n_{stk}$$

- w_{core} — radial wall width; h_{stk} — stacked core height [mm]

$$(E25) \quad MLT = 2(w_{core} + h_{stk}) + b_w \quad [\text{mm}] \quad (\text{engine})$$

- b_w — winding build allowance [mm] (default 3.8)
- Viewer form: $MLT = 2(w_{core} + h_{stk}) + 2 \cdot OD_b$, leads added once: $l = N \cdot MLT + l_{lead}$

$$(E26) \quad I_{Cu} = N \cdot MLT / 1000 \quad [\text{m}]$$

- I_{Cu} — total conductor length

$$(E27) R_{DC,25} = \rho_{20}(1 + 5\alpha_{Cu}) \cdot I_{Cu} / (A_{Cu} \times 10^{-6}) \text{ } [\Omega]$$

- geometry estimate at 25 °C; provenance ‘computed’

Note: Measured override (T3): $R_{DC,25} = R_{meas,20}(1+5\alpha_{Cu})$ or $R'_{meas,20} \cdot I_{Cu}(1+5\alpha_{Cu})$; a $>2\times$ disagreement with the geometry estimate raises a validation warning.

$$(E28) R_{DC}(T) = R_{DC,25} \cdot (1 + \alpha_{Cu}(T - 25))$$

- T — copper temperature [°C]; losses evaluated at $T_{wind} = 100$ °C (worst) and 25 °C

$$(E29) FF_{Cu} = N \cdot A_{Cu} / W_a$$

- FF_{Cu} — bare-copper window fill; W_a — core window area [mm²]; $FF_{Cu} > 1$ is a hard validation error (physically impossible)

S7 — Skin depth and AC resistance

$$(E30) \delta = \sqrt{(\rho_{20} / (\pi \cdot f_{sw} \cdot \mu_0))} \text{ } [m]$$

- δ — copper skin depth at f_{sw} ; strand check: $d_{str} < 2\delta$

$$(E31) F_R = R_{AC}/R_{DC} = 1.15 \text{ (engine default, package-overridable)}$$

- F_R — AC-to-DC resistance ratio at f_{sw} (Tier-1 stand-in heuristic)

$$(E32) \text{ Viewer analytic: } x = d_{str}/(2\delta); F_{skin} = 1 + k_{skin}x^2; F_{prox} = 1 + k_{prox}(N_{lay}-1)x^2(1 + k_{crowd}(crowd_{ax}-1)); F_R = F_{skin} \cdot F_{prox}$$

- k_{skin} , k_{prox} , k_{crowd} — stand-in hyperparameters (defaults 0.5/0.40/0.25)
- N_{lay} — bore winding layers (E55); $crowd_{ax} = r_{mean}/r_{in}$

Note: Both forms are Tier-1 placeholders. A solved table fields.windingAC{f, R_{AC}/R_{DC} } (eddy-current FEA or core-less-bobbin measurement) replaces them by interpolation at f_{sw} (T2/T3).

$$(E33) k_{harm} = \sum_{n \text{ odd}} (1/n^2)^2 n^2 / \sum_{n \text{ odd}} (1/n^2)^2 \approx 1.213$$

- k_{harm} — excess factor for the triangular-ripple harmonic spectrum ($n = 1,3,5,\dots,199$)

S8 — RMS decomposition and copper loss

$$(E34) I_{rms,LF} = I_c/\sqrt{2}$$

- low-frequency (line) RMS of the per-phase current

$$(E35) I_{rms,HF} = \sqrt{[(1/\pi) \int_0^\pi (\Delta I_{pp}(\theta)/(2\sqrt{3}))^2 d\theta]}$$

- RMS of the triangular switching ripple, cycle-averaged

$$(E36) I_{\phi,rms} = \sqrt{(I_{rms,LF}^2 + I_{rms,HF}^2)}$$

- total per-phase RMS current

$$(E37) P_{Cu}(T) = I_{\phi,rms}^2 \cdot R_{DC}(T) + I_{rms,HF}^2 \cdot (F_R - 1) \cdot k_{harm} \cdot R_{DC}(T)$$

- first term — all current at DC resistance; second — HF ripple AC excess only
- reported at T = 100 °C ($P_{Cu,100}$, worst) and 25 °C ($P_{Cu,25}$)

$$(E38) J = \max_{\text{corners}} (I_{\phi,rms}) / A_{Cu} \text{ } [A/mm^2]$$

- J — copper current density, checked against acceptance.J_max if supplied

S9 — Core loss (Steinmetz with duty correction, cycle-averaged)

$$(E39) P_v(B, f) = a \cdot B^b \cdot f_{kHz}^c \text{ } [mW/cm^3]$$

- a, b, c — material Steinmetz coefficients (catalog or measured refit); B in T, f in kHz

$$(E40) I_{c,int} = \int_0^{2\pi} |\cos\theta|^c d\theta \text{ } (\approx 3.54 \text{ for } c \approx 1.4)$$

- iGSE-style normalization integral, computed numerically per material

$$(E41) F_D(D) = 2^c \cdot (D^{1-c} + (1-D)^{1-c}) / ((2\pi)^{c-1} \cdot I_{c,int})$$

- F_D — duty-ratio (non-sinusoidal) correction for the asymmetric triangular flux; D clipped to $[10^{-4}, 1-10^{-4}]$

$$(E42) p_{core}(\theta) = P_v(B_{ac,pk}(\theta), f_{sw,kHz}) \cdot F_D(D(\theta)) \cdot V_{e,cm3} / 1000 \text{ } [W]$$

- $V_{e,cm3} = V_{e,single} \cdot n_{stk}$ — total core volume [cm³]

(E43) $P_{\text{core}} = (1/\pi) \int_0^\pi p_{\text{core}}(\theta) d\theta$ (method 'cycle_avg_igse')

- alternative 'peak_point': $P_{\text{core}} = P_v(B_{\text{ac,crest}}, f_{\text{kHz}}) \cdot V_{\text{e,cm}^3} / 1000$

(E44) $P_{\text{core,max}} = P_{\text{core}} \cdot s_{\text{max}}$

- s_{max} — material max/typical loss band (catalog default $1.286 \approx \pm 29\%$); a measured refit may supply its own bench-uncertainty s_{max}

Note: Measured core loss (T3): either a refit {a,b,c} used directly in E39, or ≥ 3 raw bench points fitted by least squares (E59); the catalog transcription anchor is then dropped, and measured-vs-catalog disagreement $> 1.5\times$ raises a validation warning.

(E45) $P_{\text{tot,typ}} = P_{\text{core}} + P_{\text{Cu,100}}, P_{\text{tot,max}} = P_{\text{core,max}} + P_{\text{Cu,100}}$

- per-corner totals; the worst corner over all operating points is reported

S10 — Thermal (engine correlation; viewer 2-node network; FEA/CFD override)

Engine (single-node correlation, T1)

(E46) $OD_w = OD + 2\sqrt{A_{\text{bund}}}, OH = h_{\text{stk}} + 2\sqrt{A_{\text{bund}}} + 6.6, \text{hole} = \max(ID - 2\sqrt{A_{\text{bund}}}, 1) \text{ [mm]}$

- $A_{\text{bund}} = n_{\text{str}} \cdot A_{\text{str}}$ — one bundle's copper area [mm²]; wound-part envelope dimensions

(E47) $SA = [\pi \cdot OD_w \cdot OH + (\pi/2)(OD_w^2 - \text{hole}^2) + \pi \cdot \text{hole} \cdot OH] / 100 \text{ [cm}^2\text{]}$

- SA — exposed surface area of the wound part

(E48) $\Delta T_{\text{nat}} = (P_{\text{tot,max}} \cdot 1000 / SA)^{0.833} \text{ [K]}$

- natural-convection correlation (P in W $\rightarrow$ mW in the ratio)

(E49) Forced: $Re = v \cdot L_c / \nu, Nu = 0.466 \cdot Re^{0.5} \cdot Pr^{1/3}, h = Nu \cdot k_{\text{air}} / L_c, h_{\text{rad}} = 0.9 \cdot \sigma \cdot 4 \cdot T_m^3, \Delta T = P / (SA_{\text{m}2}(h + h_{\text{rad}}))$

- v — airflow [m/s]; $L_c = OD_w / 1000 \text{ [m]}$; $\nu = 1.57 \times 10^{-5} \text{ m}^2/\text{s}$; $Pr = 0.71$; $k_{\text{air}} = 0.0263 \text{ W/m}\cdot\text{K}$; $\sigma = 5.67 \times 10^{-8}$; T_m — mean film temperature [K]

Viewer (two-node core/winding network, T1; calibrated to E42)

(E50) $\theta_{\text{sa}} = \Delta T_{\text{nat}} \cdot c_{\text{mult}} / P_{\text{tot}}; R_{\text{ca}} = \theta_{\text{sa}}(s_{\text{c}} + s_{\text{w}}) / s_{\text{c}}, R_{\text{wa}} = \theta_{\text{sa}}(s_{\text{c}} + s_{\text{w}}) / s_{\text{w}}, R_{\text{cw}} = \theta_{\text{sa}} \cdot c_{\text{cw}}$

- $s_{\text{c}}, s_{\text{w}}$ — core/winding split of the ambient path ($1/R_{\text{ca}} + 1/R_{\text{wa}} = 1/\theta_{\text{sa}}$); $c_{\text{mult}}, c_{\text{cw}}$ — cooling-model factors from the package

(E51) Steady state: $[P_{\text{core}}; P_{\text{Cu}}] = [1/R_{\text{ca}} + 1/R_{\text{cw}}, -1/R_{\text{cw}}; -1/R_{\text{cw}}, 1/R_{\text{wa}} + 1/R_{\text{cw}}] \cdot [\Delta T_{\text{c}}; \Delta T_{\text{w}}]$

- 2×2 linear solve for core and winding rises; $T_{\text{hot}} = T_{\text{amb}} + \max(\Delta T_{\text{c}}, \Delta T_{\text{w}}) \cdot k_{\text{hot}}$

(E52) Warm-up: $C_{\text{th,c}} dT_{\text{c}}/dt = P_{\text{core}} - (T_{\text{c}} - T_{\text{a}})/R_{\text{ca}} - (T_{\text{c}} - T_{\text{w}})/R_{\text{cw}}$ (and symmetrically for the winding), Euler $dt = 0.5 \text{ s}$

- $C_{\text{th,c}}, C_{\text{th,w}}$ — node thermal capacitances [J/K] from the package

(E53) FEA/CFD override (T2): $\{R_{\text{ca}}, R_{\text{wa}}, R_{\text{cw}}\}$ from fields.thermal.nodes; engine winding rise $\Delta T = P_{\text{tot,max}} \cdot R_{\text{wa}}; \theta_{\text{SA}}$ -only form expands as $R_{\text{ca}} = R_{\text{wa}} = 2\theta_{\text{SA}}, R_{\text{cw}} = \theta_{\text{SA}}/2$

S11 — Window utilization, inductance guarantee, DCM boundary

(E54) $OD_b = 2\sqrt{(A_{\text{Cu}} / FF_b / \pi)} \text{ [mm]} \text{ (single strand: } 1.08 \cdot d_{\text{str}})$

- OD_b — effective bundle diameter; FF_b — bundle packing factor (default 0.55)

(E55) $t_{\text{pl}} = \lceil 2\pi \cdot \max(r_{\text{in}} - OD_b/2, OD_b) / OD_b \rceil, N_{\text{lay}} = \lceil N/t_{\text{pl}} \rceil, r_{\text{bore}} = r_{\text{in}} - N_{\text{lay}} \cdot OD_b$

- t_{pl} — turns per bore layer; N_{lay} — layers needed; r_{bore} — residual bore radius (negative $\Rightarrow$ overfull)

(E56) $K_u = N \cdot A_{\text{Cu}} / (\pi \cdot r_{\text{in}}^2)$

- K_u — bore-window utilization, checked against acceptance. K_{u_max} if supplied

(E57) $L_{\text{guar}} = \min_{\text{corners}} L_{\text{min}}(H_{\text{pk}})$; analytic: $L_{\text{min}}(H) = k(H) \cdot L_{0,\text{min}}$; table: un-derated interp

- $H_{\text{pk}} = 0.4\pi \cdot N \cdot I_{\text{pk,inst}} / I_{\text{e,cm}}$ — bias at the worst instantaneous peak
- checked against acceptance.L_target_uH if supplied

(E58) DCM boundary: load where $I_c = \Delta I_{\text{pp}}(\text{crest})/2$; waveform flag dcm(θ) when $i_{\text{avg}}(\theta) < \Delta I_{\text{pp}}(\theta)/2$

- below the boundary the CCM equations are invalid; the viewer shades the region and warns when the DCM fraction of the half-cycle exceeds 0.5

S12 — Worst-case scan and acceptance verdict

All operating corners (engine) / all envelope V_{in} at spec-max load (viewer) are evaluated; the scan reports: worst $P_{\text{tot,max}}$ and its V_{in} ; worst B_{max} (mean path) and B_{inner} ; minimum L_{guar} ; maximum ΔT ; maximum J ; K_u . Acceptance checks ONLY limits present in the package (upstream-only — omitted limit $\Rightarrow$ 'no upstream limit', no value invented):

- material anchors reproduce catalog points (skipped for measured fits) • $L_{\text{guar}} \geq L_{\text{target}}$ • $B_{\text{max}} \leq B_{\text{max,T}}$ or $(1 - \text{sat_margin_min}) \cdot B_{\text{sat}} \cdot B_{\text{inner}} \leq B_{\text{inner_max_T}}$ (optional) • $\text{FF}_{\text{Cu}} \leq \text{FF}_{\text{Cu_limit}}$ • $d_{\text{str}} < 2\delta$ • $\Delta T \leq T_{\text{hot}} - T_{\text{amb}}$ (or $dT_{\text{max_K}} \cdot J \leq J_{\text{max}}$. Verdict = APPROVE iff every checked assert passes; REJECT otherwise; NO LIMITS / INSUFFICIENT DATA when nothing checkable.

S13 — Measured-data (Tier-3) equations

(E59) Steinmetz refit: $\ln P_i = \ln a + b \cdot \ln B_i + c \cdot \ln f_i \rightarrow$ least-squares solve for ($\ln a$, b , c) over ≥ 3 bench points

- points [B_i (T), f_i (kHz), P_i (mW/cm³)] must span B and f

(E60) Bias-current conversion: $H_i = 0.4\pi \cdot N \cdot I_i / I_{\text{e,cm}}$; points sorted ascending before interpolation

- lets the lab supply L vs I directly; engine/viewer convert with the design's own N and I_{e}

(E61) Self-consistency guards (warn, never silently accept): measured Steinmetz vs catalog ratio at (0.1 T, 100 kHz) outside [0.67, 1.5]; median bench-point ratio outside [0.67, 1.5]; measured L_0 vs analytic L_0 off by $>25\%$; measured R_{DC} vs geometry off by $>2\times$

Appendix — Variable glossary

Symbol	Package / code name	Unit	Meaning
mu0	MU0	H/m	Permeability of free space ($4\pi \cdot 10^{-7}$)
rho20	copper.rho20_ohm_m	ohm*m	Copper resistivity at 20 C
alphaCu	copper.alphaCu	1/K	Copper temperature coefficient
Vout	design.Vout	V	Boost DC output voltage
fsw	design.fsw	Hz	Switching frequency
fline	design.lineHz	Hz	AC line frequency
nph	design.nph	-	Interleaved phase count
Prated	design.Prated	W	Rated output power (map-derived points)
Vin, Pout, eta, PF	operating.points[i]	V, W, -, -	Spec-corner line voltage, output power, efficiency, PF
Tamb	environment.Tamb_C	C	Worst-case ambient temperature
Thot	environment.Thot_C	C	Allowed hotspot temperature
N	winding.N	turns	Number of turns
nstk	winding.stacks	-	Stacked core count
bw	winding.build_mm	mm	Winding build allowance for MLT
OD, ID, HT	geometry.OD/ID/HT_mm	mm	Core outer/inner diameter, SINGLE-core height
Ae,single	geometry.Ae_mm2	mm ²	Effective area of ONE core
Le	geometry.Le_mm	mm	Magnetic path length ($l_{e_cm} = Le/10$)
Ve,single	geometry.Ve_mm3	mm ³	Volume of ONE core
Wa	geometry.Wa_mm2	mm ²	Window area
AL,single	geometry.AL_nH	nH/T ²	Inductance factor of ONE core, zero bias
tolAL	geometry.AL_tol	-	AL tolerance fraction (analytic derate only)
Bsat	material.Bsat	T	Saturation flux density
a, b, c	material.steinmetz	-	Core-loss fit $P = a \cdot B^b \cdot f_{kHz}^c$ (mW/cm ³)
k0, k1, p	material.retention	-	DC-bias fit $\mu = 1/(k_0 + k_1 \cdot H_{Oe}^p)$
smax	material.lossMaxScale	-	Max/typ core-loss band (default 1.286)
dstr	copper.wire.strandDia_mm	mm	Bare strand diameter
nstr, npar	wire.strands / parallel	-	Strands per bundle / parallel bundles
FFb	wire.fillFactor	-	Bundle packing factor (bundle OD)
kskin,kprox,kcrowd	copper.prox	-	Viewer analytic R_AC stand-in hyperparameters
FR	copper.RacRdc / fields.windingAC	-	AC/DC resistance ratio at fsw
Rmeas20	copper.measured.RdcTotal_20C	ohm	Measured total DCR at 20 C (T3)
theta	(internal)	rad	Line half-cycle angle, grid over (0, pi)
D	(internal)	-	Boost duty ratio $1 - v_{in}/V_{out}$
Ic	(internal lphi_crest)	A	Per-phase crest current
dlpp	(internal)	A	Peak-to-peak switching ripple
H	(internal)	Oe	Bias field $0.4 \cdot \pi \cdot N \cdot l / l_{e_cm}$
k(H)	(internal)	-	Permeability retention at bias H
L0 / L(H)	statics.L0_*_uH / points.L_nom_uH	H, uH	Unbiased / bias-corrected inductance
Bac,pk	points.Bac_crest	T	Half of pk-pk AC flux swing
Bmax	worst.Bmax.Bmax	T	Worst mean-path instantaneous peak flux
Binner	worst.Binner.B_inner	T	Crowded inner-radius peak flux
crowd	statics.crowd_peak	-	Inner-radius crowding ratio r_{mean}/r_{in} (or FEA)

Symbol	Package / code name	Unit	Meaning
MLT	statics.MLT_mm	mm	Mean length per turn
lCu	statics.ell_cu_m	m	Total conductor length
ACu	(internal A_cu_mm2)	mm2	Total copper cross-section
RDC(T)	statics.DCR25/100_mohm	mohm	DC resistance at temperature T
delta	statics.skin_depth_mm	mm	Skin depth at fsw
kharm	(internal ~1.213)	-	Harmonic excess factor, triangular ripple
Irms,LF/HF	points.lphi_rms / Irms_HF	A	Line / ripple RMS components
PCu	points.Pcu25/Pcu100	W	Copper loss at 25/100 C
Fd(D)	(internal F_D)	-	Duty correction for core loss
Pcore(,max)	points.Pcore / Pcore_max	W	Cycle-avg core loss, typ / max band
Ptot	points.Ptot_typ/_max	W	Total loss per corner
SA	statics.SA_cm2	cm2	Wound-part surface area
dT	points.dT / worst.dT	K	Temperature rise (correlation, nodes, or CFD)
Rca,Rwa,Rcw	fields.thermal.nodes	K/W	Core-amb / winding-amb / core-winding resistances
ODb	windowGeom.bundleOD	mm	Effective bundle diameter
Nlay, tpl	windowGeom.layersNeeded/turnsPerLayer		Bore layers / turns per layer
Ku	windowGeom.Ku	-	Bore utilization $N \cdot ACu / (\pi \cdot r_{in}^2)$
J	statics.J_AperMM2	A/mm2	Copper current density
Lguar	worst.Lmin_guarantee_uH	uH	Guaranteed minimum inductance at worst bias
FFCu	statics.FFCu	-	Bare-copper window fill $N \cdot ACu / W_a$
provenance/tier	result.provenance / tier	-	Per-quantity source: input-T0, analytic-T1, fea-T2, measured-T3

*Honest status note: F_R (E31/E32), the thermal correlations (E46–E52), and the analytic L(H) (E7–E8) are Tier-1 stand-ins. They are replaced — per quantity, with provenance — when solved FEA/CFD data arrives in **fields** or measured bench data arrives in **material.measured** / **copper.measured**. Everything else above is exact arithmetic on the injected package values.*